identical length, sampled from proteins that do not contain the motif. All proteins and domain locations were retrieved through the UniProt API [1]. This yields a balanced dataset of local contexts that do or do not contain the target motif.

Window embedding We then create embeddings for each positive and negative sequence using a Protein Language Model, in this case ESM-C [5] Each protein subsequence is represented as an ordered sequence of amino acids

$$S = (a_1, a_2, \ldots, a_n), \quad a_i \in \mathcal{A},\ |\mathcal{A}| = 20,$$

where n is the subsequence length.

Passing S through a pretrained protein language model yields contextualized residue embeddings. For a chosen model layer m, the activations are expressed as

$$H_m = \begin{bmatrix} h_{1m} \\ h_{2m} \\ \vdots \\ h_{nm} \end{bmatrix} \in \mathbb{R}^{n \times d},$$

where $h_{im} \in \mathbb{R}^d$ denotes the d-dimensional embedding of residue a_i.

To obtain a fixed-size representation for each window, we must summarize its residue embeddings to a single vector. Several pooling strategies exist for this purpose, including CLS-Pooling, Mean-Pooling, and the recently proposed BoM-Pooling [8]. In this work, we use Mean-Pooling, which computes the average embedding over all residues to obtain a single global representation.

$$\bar{h}_m = \frac{1}{n} \sum_{i=1}^{n} h_{im}.$$

Despite its simplicity, Mean-Pooling is consistently effective in practice [18]. As motifs are local in nature, we produce an embedding for each individual subsequence, rather than embedding the whole sequence and then subsetting these embeddings to windows. Recent analyses of ESM models demonstrate that structural signals, such as residue–residue contacts, are encoded through local sequence windows, with even short stretches of sequence sufficient to recover predicted contacts [21]. We therefore embed each window independently, to better capture local motif-associated signals without interference from distant regions of the protein.

Learning the Functional Direction. Given pooled window embeddings for positive and negative samples for a target motif, we then train a linear classifier (logistic regression) to distinguish between the two distributions (Figure 1B). Let the separating hyperplane be defined as

$$w^\top x + b = 0,$$

where x is a d-dimensional window embedding. The normal vector w orthogonal to this boundary, oriented toward the positive class, captures the direction in embedding space that best characterizes the motif. We interpret this vector as the motif's *Concept Activation Vector (CAV)*[10], representing the direction in $\mathbb{R}^m$ that captures how strongly the motif concept is expressed in the embedding space.

Training is performed independently for each motif, producing a dictionary of motif vectors:

$$\mathcal{V} = \{v_1, v_2, \ldots, v_K\}, \quad v_k \in \mathbb{R}^d,$$

where each v_k corresponds to a learned motif concept. This stage effectively converts qualitative biochemical properties (e.g., "zinc-finger-ness") into quantitative geometric directions within the model's representation space. Prior work has shown that neural embedding spaces can encode semantic relations as approximately linear directions, with relation-specific vector offsets supporting vector-style reasoning in the embedding space [14]. In our framework, motifs are represented as interpretable directions in embedding space, and these motif vectors can be directly used to score new sequences, identifying regions that align strongly with a given motif concept.

Stage II: Motif Localization via Concept Alignment

Window-Level Embedding Extraction. To detect and localize functional motifs within proteins, we embed short windows of subsequences (Figure 1C).

Formally, every window of length w beginning at position t is fed to the pretrained model as a standalone sequence,

$$S_{t:t+w}.$$

The window size w is motif-specific: for each Pfam family, we compute the median annotated length from the training data and add a small buffer to account for natural variation across homologs. This yields motif-appropriate receptive fields without excessively enlarging the search space. We slide this window with a stride set to one-half of its length, ensuring dense coverage and allowing overlapping windows to compete for high alignment scores. Such overlap is essential, since true motif boundaries rarely align perfectly with a fixed grid and often require multiple shifted windows to capture their maximal signal.

Its internal activations $H_m^{(t)} \in \mathbb{R}^{w \times d}$ are then pooled to produce a single representation,

$$\bar{h}_{t,m} = \frac{1}{w} \sum_{i=1}^{w} h_{t,m}^{(i)}.$$

Computing Concept Alignment Scores. Given the window embedding $\bar{h}_{t,m}$ and the learned CAV v_k corresponding to motif k, we measure the degree of concept alignment through the inner product:

$$s_{k,t} = \langle v_k, \bar{h}_{t,m} \rangle.$$

This alignment score reflects how closely the representation of the current window expresses the functional motif concept. Higher scores indicate stronger motif-like signal in that region. We thus rank candidate motif locations in the protein by CAV score.

Code Availability

Relevant code to perform analyses is available at https://github.com/A-Shamail/TCAV/.

3. Results

Choice of Layer for CAV Scoring. We first examine how the choice of model layer affects the quality of the CAV signal. Figure 2A shows smoothed CAV score profiles produced by all 35 layers of ESM-C on the example protein. Each amino acid is assigned the mean alignment score of all windows in which it appears, producing a continuous curve that reflects how strongly each motif is expressed at every position. Although all layers follow the same broad trend, each responding to the same underlying motif structure, the magnitude, sharpness, and signal-to-noise ratio differ substantially across the network.

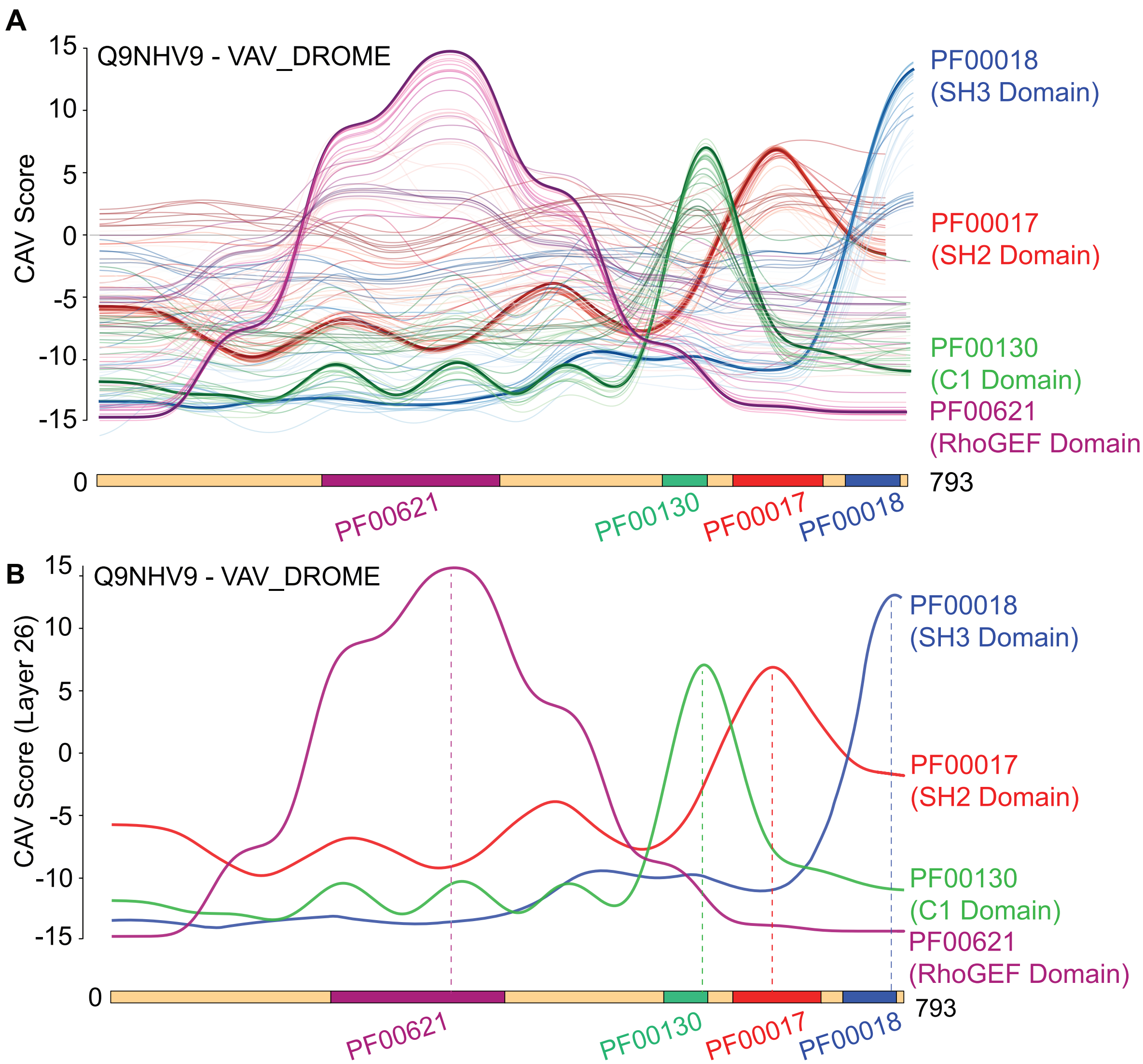

Figure 2: (A) Layerwise CAV score profiles for all 36 layers of ESM-C 600M. Bold curves denote CAV scores for layer 26 (indexed from 1). Excluding the bold curves, earlier layers are lighter and later layers are darker. (B) CAV peaks fall in ground-truth domain intervals.

A consistent pattern emerges: early layers exhibit weak motif-specific signal, mid-level layers (approximately layers 21–31) produce the strongest and most discriminative peaks, and the deepest layers (32-36) again show diminished performance. This symmetric behavior suggests that motif-related information is represented most cleanly in the middle layers, where local biochemical features and longer-range dependencies are jointly available but not yet saturated by high-level abstraction. This observation is consistent with a broad body of representation-learning research showing that middle layers of deep networks concentrate semantically meaningful structure, while early layers